

MSDS 535 – Week 16

A.3, A.4

Overview and Objectives

- Matrix and linear algebra.
- Concepts and tools from signal processing.

Linear System $Ax = b$

- Standard square systems are solvable using the matrix inverse A^{-1} when A is invertible.
- If A is not invertible, alternative methods like Gaussian elimination or LU decomposition may be employed.
- Iterative methods, such as the Jacobi or Gauss-Seidel method, provide solutions by iterating towards convergence.
- Preconditioning techniques involve transforming the system to a more suitable form for faster solving.

Overdetermined System

- Least square solutions aim to minimize the residual errors between Ax and b for systems with more equations than unknowns.
- The Moore-Penrose pseudo-inverse is utilized for non-square matrices or cases where A^TA is not invertible.
- Techniques like QR decomposition or SVD often help in handling overdetermined systems efficiently.

Overdetermined System

- Ridge regression with an l_2 penalty term helps control solutions and handles multicollinearity issues.

Underdetermined System

- Underdetermined systems have infinite solutions, and the least square solution selects the one with the smallest norm.
- Methods like basis pursuit or compressed sensing are employed to find sparse solutions in underdetermined systems.

Underdetermined System

- Regularization techniques such as Tikhonov regularization aid in controlling the solution space.
- Application-specific constraints or additional information may help narrow down solutions in practical scenarios.

Norms of Vectors and Matrices

- Norms measure the size or length of vectors and matrices in different ways, reflecting various properties.
- Matrix norms, including the Frobenius norm, spectral norms, or induced norms, have specific applications.

Norms of Vectors and Matrices

- Vector norms like the L_1 , L_2 , and L_∞ norms offer different ways to measure vector magnitudes.
- Dual norms and their relationships are essential in understanding optimization problems.

Matrix Decompositions

- Eigendecomposition diagonalizes matrices, revealing eigenvalues and eigenvectors, particularly useful for symmetric matrices.
- Singular Value Decomposition (SVD) decomposes matrices into orthogonal and diagonal matrices, applicable universally.

Matrix Decompositions

- QR decomposition breaks matrices into orthogonal and upper triangular matrices, commonly used in solving linear equations.
- LU decomposition separates matrices into lower and upper triangular matrices, simplifying matrix inversion.

Singular Value Decomposition (SVD)

- SVD finds applications in data compression, signal processing, and image reconstruction.
- Truncated SVD and low-rank approximations help in reducing the dimensionality of data while preserving essential information.
- Principal Component Analysis (PCA) relies on SVD to identify essential patterns and reduce data complexity.
- Applications in recommendation systems and collaborative filtering showcase the versatility of SVD.

Subspace

- Subspaces are defined by linear combinations of vectors, providing insights into solution spaces.
- Null spaces and column spaces reveal crucial properties about a matrix's kernel and range, respectively.
- Orthogonal complements help understand the relationship between subspaces and their orthogonal spaces.
- Connections between subspaces and their orthogonal complements are vital in decomposition theorems.

Orthogonality

- Orthogonal vectors form a basis in Euclidean spaces, providing simplicity in calculations.
- Orthonormal bases facilitate orthogonal projections and simplifications in functional analysis.
- Orthogonal transformations preserve inner products, ensuring angle and distance preservation.
- Orthogonal matrices play a vital role in rotations, reflections, and transformations in various applications.

Entropy

- Entropy, a measure in information theory, quantifies the uncertainty associated with a random variable.
- The entropy of a discrete random variable X , denoted $H(X)$, is determined by its probability mass function $p_X(x)$.
- Differential entropy characterizes uncertainty in a continuous random variable Y using its probability density function $p_Y(y)$.

Entropy

- Entropy, reflected in $H(X)$ and $H(Y)$, gauges the unpredictability or information content in the respective distributions.
- Key properties include non-negativity and the chain rule ($H(X,Y) = H(X) + H(Y | X)$).

Other commonly used entropy measures

- Joint Entropy, $H(X,Y)$, assesses uncertainty in jointly distributed random variables (X,Y) using their joint probability distribution.
- Conditional Entropy, $H(X|Y)$, quantifies the information content of X given knowledge of Y , considering joint and marginal distributions.
- Fano's inequality offers insights into incorrect estimations in a Markov chain scenario, linking estimation certainty with variable uncertainty.

Kullback-Leibler divergence (KL-divergence)

- KL-divergence measures the discrepancy between two probability distributions.
- It remains nonnegative and equals zero only if the distributions are identical.
- Lacks symmetry and metric properties but follows the chain rule and exhibits convex behavior regarding probability mass functions.

Mutual information

- Mutual information, $I(X; Y)$, measures the information gained about one random variable by observing another.
- Properties include non-negativity, symmetry, and connections with entropy via conditional entropy and joint entropy.
- Follows the chain rule and adheres to the data processing inequality, guiding information transfer between variables.

Discrete Fourier transform (DFT) and fast Fourier transform (FFT)

- Finite sequences and their relation to periodic sequences lay the groundwork for understanding DFS (Discrete Fourier Series).
- DFT of finite sequences showcases linearity, circular shift properties, and the orthogonality of basis vectors.

Discrete Fourier transform (DFT) and fast Fourier transform (FFT)

- Examining properties of DFT, including Parseval's theorem, duality, and symmetry, becomes crucial in signal processing and analysis.
- Introducing Fast Fourier Transform (FFT), a set of algorithms optimizing DFT computation in diverse applications, revolutionizing fields like mathematics, engineering, and music.